

Documenting Changes in 01_download/README.Rmd

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Purpose

Provide a human-readable summary of all download scripts in R/01_download/, including their purpose, input/output expectations, status, and key changes over time. This supports reproducibility and onboarding of new contributors.

Overview

This section documents the current structure and status of all scripts responsible for downloading raw spatial and tabular data inputs. Each script is version-controlled and tied to a specific covariate or source dataset.

Table 1: Script Index

File Name	Description	Output Folder	Notes / Status
<code>_latex_preamble.tex</code>	CHECK		
<code>01a_download_ecoregions.R</code>	Download Level I-IV US ecoregion shapefiles	<code>data/raw/</code>	CRS standardized (EPSG:5070)
<code>01b_download_usgs_gage_data.R</code>	Download peak flow gage locations	<code>data/raw/</code>	Reorganized metadata
<code>01c_download_usgs_gage_metadata.R</code>	Detailed site metadata	<code>data/raw/</code>	Outputs raw data-dictionary
<code>01d_download_usgs_peakflow_data.R</code>	Download annual peak flow values	<code>data/raw/</code>	Handles chunking & retries
<code>01e_filter_usgs_peakflow_data.R</code>	Filter data, calculate skew, tag PILFs	<code>data/processed/</code>	Replaces legacy script
<code>01f_download_nhdplus_v21.R</code>	Download NHDPlusV2.1 flowlines and catchments	<code>data/processed</code>	CRS standardized (EPSG:5070)
<code>01g_download_nhdplus_hr_catchments.R</code>	Download NHDPlus HR (1:24k) catchment boundaries	<code>data/raw</code>	Idempotent, Geometry-safe, Fuzzy retry logic: name correction and tile, Transparent logging
<code>01h_download_nhdplus_hr_flowlines.R</code>	Download NHDPlus HR (1:24k) flowlines	<code>data/raw</code>	Logs success/failure, retry logic, diagnose geometry issues
<code>01j_download_koppen-geiger_data.R</code>	Download and process Köppen-Geiger climate raster	<code>data/processed</code>	CRS standardized (EPSG:5070)

File Name	Description	Output Folder	Notes / Status
01k_download_plant-hardiness.R	Download and process USDA Plant Hardiness Zone Map (PHZM)	data/processed	CRS standardized (EPSG:5070)
01l_download_prism_climate.R	Download and process PRISM climate normals	data/processed	CRS standardized (EPSG:5070)
01m_download_nlcd_2016.R	Download NLCD 2016 Land Cover raster clipped to Great Plains	data/processed	CRS standardized (EPSG:5070)
01n_download_ned.R	Download NED (National Elevation Dataset) clipped to Great Plains and calculate slope	data/processed	CRS standardized (EPSG:5070)
01o_download_modis_2016.R	Document the process for downloading MODIS MOD13Q1 (NDVI/EVI), then reproject and clip to Great Plains	data/processed	CRS standardized (EPSG:5070)
01p_download_statsgo2.R	Download STATSGO2 (national soil database) mapunit data	data/processed	clipped to GP Ecoregion using Soil Data Access (SDA). CRS standardized (EPSG:5070)

Summary of Key Changes

Standardized Folder Paths

- All output paths now use `here::here("data", "raw", ...)` or `data/processed/`
- Ensures reproducibility across systems and OSes

Consistent CRS

- All spatial outputs reprojected to EPSG:5070 (CONUS Albers Equal Area)

Modular Design

- Each script targets a single source or covariate
- Utility functions moved to `R/utils/`

Logging & QA

- Retry attempts, download failures, and chunking behavior logged to `.csv` where applicable
 - Scripts now include `{cli}` alerts for status tracking
 - Each script is cross-linked to relevant milestone reports
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